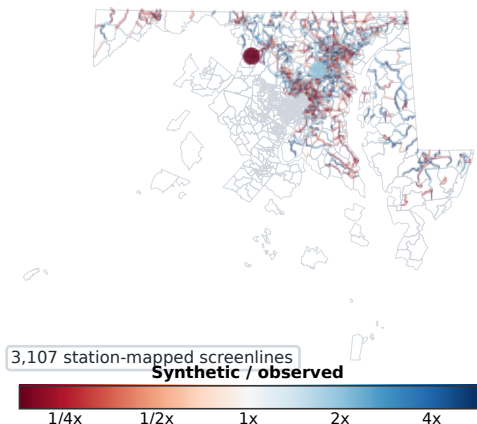


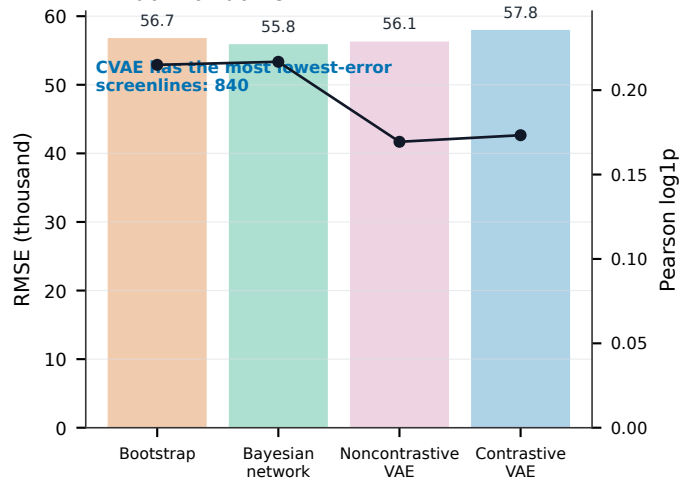
External screenline validation

Annual tract-boundary screenlines and hourly TMAS cells test whether synthetic (b) support remains externally plausible.

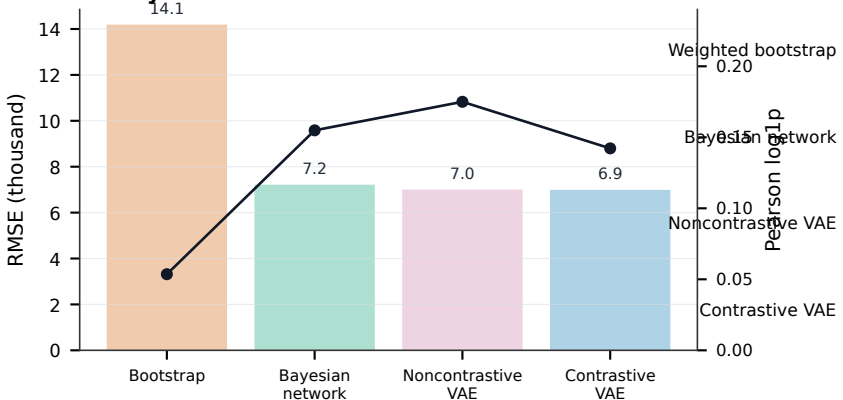
(a) Annual screenline ratio, Contrastive VAE



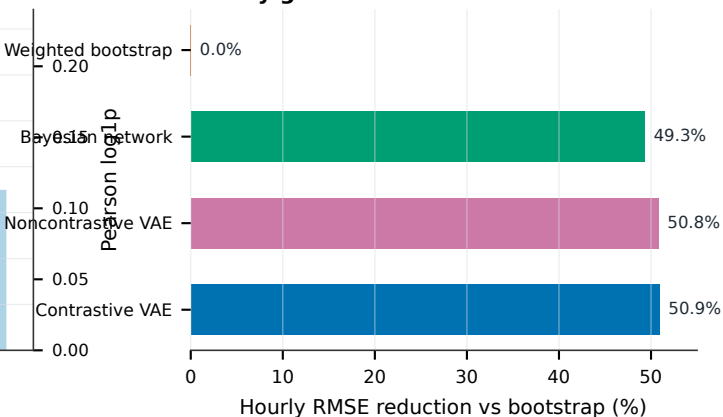
(b) Annual validation



(c) Hourly TMAS validation



(d) Hourly gain



Screenline counts are geometric tract-boundary proxy comparisons, not routed network assignments.